

Employee Attendance Management System-Setup Guide

Frontend Setup:-

Step 1: Create the React + Vite App

Open your terminal in the root project folder and run:

Terminal :

```
npm create vite@latest client -- --template react
```

Step 2: Navigate into the Client Directory

Change into the newly created frontend directory:

Terminal:

```
cd client
```

Step 3: Installed Packages (Dependencies Breakdown)

- Core & Production Dependencies:
 - react & react-dom — Core UI library.
 - react-router-dom — Client-side routing, protected routes, and redirects.
 - axios — HTTP client for API requests and JWT authorization headers.
 - lucide-react — Icon set (e.g., Mail, Lock, User, Logout, AlertCircle, Loader2).
- Developer & Styling Dependencies:
 - vite — Next-generation frontend build tool and dev server.
 - tailwindcss — Utility-first CSS styling framework.
 - postcss & autoprefixer — CSS post-processing engine.

Step 4: Commands to Install All Packages

Run the following commands sequentially inside the client directory:

1. Install Vite & React Base Dependencies:

Terminal:

```
npm install
```

2. Install Additional Production Dependencies:

Terminal:

```
npm install axios react-router-dom lucide-react
```

3. Install Tailwind CSS & Generate Configuration Files:

Terminal:

```
npm install -D tailwindcss postcss autoprefixer
```

```
npx tailwindcss init -p
```

Step 5: Start the Frontend Development Server

Once installation completes, launch the development server:

Terminal:

```
npm run dev
```

Backend Setup:-

STEP 1 — Backend Folder Create

Project structure:

employee-attendance-system/server

Terminal:

```
mkdir employee-attendance-system
```

```
cd employee-attendance-system
```

```
mkdir server
```

```
cd server
```

STEP 2 — Initialize Node.js Project

```
npm init -y
```

STEP 3 — Required Packages Install

```
npm install express mongoose dotenv cors bcryptjs jsonwebtoken
```

Development package:

```
npm install -D nodemon
```

STEP 4 — package.json Setup

package.json me scripts:

```
"scripts": {  
  "start": "node server.js",  
  "dev": "nodemon server.js"  
}
```

Server run:

```
npm run dev
```

STEP 6 — MongoDB Atlas Setup

MongoDB Atlas:

1. Account create
2. Free cluster create
3. Database User create
4. Username/password set
5. Network Access:
0.0.0.0/0

for development:

6. Connection string .env:

```
PORT=5000
```

```
MONGO_URI=your_mongodb_connection_string
```

```
JWT_SECRET=your_secret_key
```